LAB 10 ANSWERS REPORT

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1. Develop and demonstrate if-then-else conditional statements in Prolog via:

a) Operators and Rules.

```
% If-then-else with Operators and Rules
check_if_student_smart(X) :-
    ( X >= 85 ->
        write('The student achieved a HD.')
    ;
        write('The student did not achieve a HD.')
    ).
```

```
20 ?- check_if_student_smart(89).
The student achieved a HD.
true.
```

b) A Negation Rule.

```
% If-then-else with Negation rule
raining(monday).
raining(sunday).

is_raining(Day) :-
    ( \+ raining(Day) ->
        write("It is not raining on this day.")
    ;
        write('It is raining on this day.')
    ).
```

```
28 ?- is_raining(wednesday).
It is not raining on this day.
true.
```

c) That use Input Variables.

```
% If-then-else that use input variables
check_if_broke :-
    write('Enter how much savings you have: '),
    read(Money),
    (Money >= 1000 ->
        write('You have alot of money for a uni student.')
    ;
        write('You are broke. This is normal for a uni student.')
    ).
```

```
30 ?- check_if_broke.
Enter how much savings you have: 1738.
You have alot of money for a uni student.
true.
```

2. Develop and demonstrate an if-then-else conditional in Prolog that has multiple branches (multiple nested if-then-else statements).

```
% If-then-else with multiple branches
check_grade(Mark) :-
    (Mark >= 85 ->
        write('High Distinction')
    ; Mark >= 75 ->
        write('Distinction')
    ; Mark >= 65 ->
        write('Credit')
    ; Mark >= 50 ->
        write('Pass')
    ;
        write('Fail')
    ).
```

```
31 ?- check_grade(73).
Credit
true.
```

3. Challenge Question: Develop ‘if-then-else’ rule definitions which allows Prolog programmers to write procedural conditional statements (standard if-then-else statements), (careful not to over-write any built-in predicates).

```
% Challenge question
if_statement(Condition, Then, Else) :-
    (call(Condition) ->
        call(Then)
    ;
        call(Else)
    ).

check_age(Age) :-
    if_statement(
        Age >= 18,
        write('You are old.'),
        write('You are young.')
    ).
```

```
33 ?- check_age(21).
You are old.
true.
```